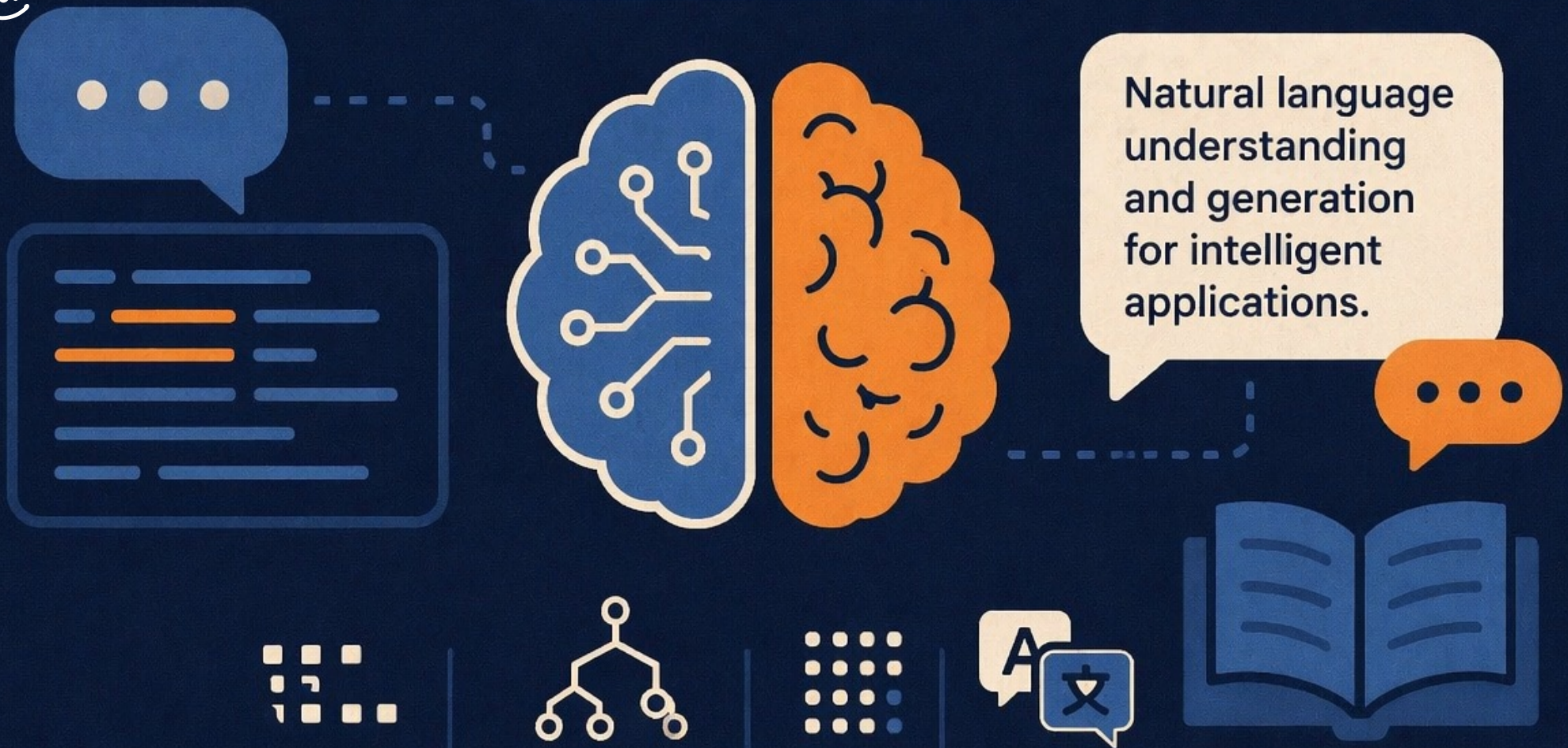
NATURAL LANGUAGE PROCESSING

class starts @ 9:05 pm



Lecture: Attention

#Agenda

1) NER Coding
2) Attention Mechanism

Attention

The coffee is very cold ?

The food was great but the movie was boring

Aspect based Sentiment Analysis

#Self Attention

K = [Hello, world, is, a, great, place]

Q = [Hello, world, is, a, great, place]

(Self attention)

User Input → $Q = [w_1, w_2, w_3, w_4 \ldots w_{100}]$

$K = [w_1', w_2' \ldots w_{100}']$

→ Text

Search Engine

Chat Bot

"positional Encoding" + AM

I loved the movie but it was too much action

↳ negation

Multi-Head AM

K = Food was great but price is high

Aspect → cost (Q)

$$AM(K, Q, V) = \begin{cases} CV_1 \\ CV_2 \\ CV_3 \\ CV_4 \end{cases}$$

S_1, S_2

↳ 10W ↳ 20W.

$K = [\ 30W\ ]$

$V = [\ 30W\ ]$

Self Attention

Self Attention

↳ Solves the idea of no needing the exact aspect work for training.


" Food was great but price is high "
high
Aspect →
Food
was
great
. .